

12. (2 points)

The Exponential Dispersion Family (“EDF”) is a family of distributions with probability density function $\pi(y; \theta, \phi)$ of the form:

$$\ln \pi(y; \theta, \phi) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)$$

and variance function of the form:

$$V(\mu) = b''((b')^{-1}(\mu))$$

a. (0.5 point)

Briefly describe the parameter θ and the function $b(\theta)$ in the equation above.

b. (0.5 point)

Explain how the Tweedie sub-family can be obtained from the EDF above.

c. (0.25 point)

Briefly describe how to select a Tweedie distribution for a heavy-tailed exposure.

Given the following information as of December 31, 2018:

Accident Year	Cumulative Paid Loss (\$000) as of (months)		
	12	24	36
2016	180	1,000	1,200
2017	200	600	
2018	150		

A GLM is used to model the incremental paid losses. The GLM uses a log link function and a Tweedie error distribution specified by:

- $p = 1$
- $b(\theta) = \exp(\theta)$
- $\mu = \exp(\theta)$
- $\ln \pi(y; \mu, \phi) = [y \ln \mu - \mu]/\phi$

d. (0.75 point)

Calculate the accident year 2018 ultimate loss.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 12	
TOTAL POINT VALUE: 2.0	LEARNING OBJECTIVE(S): A8,A9
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample Responses for parameter θ</u> • Location parameter • Canonical parameter • $\theta = (b')^{-1}(\mu)$ <u>Sample Responses for $b(\theta)$</u> • Cumulant function • $b(\theta)$ describes the mean of the EDF: $\mu = b'(\theta)$ • Shape function	
Part b: 0.5 point	
<u>Sample Responses</u> • Tweedie subfamily can be obtained by allowing the variance function to be $V(\mu) = \mu^p$, where p is not between 0 and 1. • Tweedie can be obtained from the EDF by restriction of the variance function: $V(\mu) = \mu^p$, $p \leq 0$ or $p \geq 1$	
Part c: 0.25 point	
<u>Sample Responses</u> • Choose a larger value for p • Select a Tweedie with $p = 2$, since this is the Gamma which is heavy-tailed • For heavy-tailed, use a Tweedie with $p = 3$ (inverse Gaussian)	
Part d: 0.75 point	
<u>Sample 1</u> This is an over dispersed Poisson model, if the Mack assumptions hold, which we assume they do, we can solve this using CL development factors. $12-24 = \frac{1000+600}{180+200} = 4.21; 24-34 = \frac{1200+1000+600}{1000} = 1.2$ 2018 ultimate = $4.21 \times 1.2 \times 150 = 757.89$ <u>Sample 2</u> ODP dist. $\alpha_{2016} = 1200, \beta_3 = 200/1200 = 0.167, \alpha_{2017} = 600/1 - 0.167 = 720,$ $\beta_2 = (820 + 400)/(1200 + 720) = 0.635, \alpha_{2018} = 150/(1 - 0.167 - 0.635) = 758,$ $\beta_1 = (180 + 200 + 150)/(1200 + 720 + 758) = 0.198$ 2018 Ult = $150 + 758 \times (0.635 + 0.167) = 758$	
EXAMINER'S REPORT	
Candidates were expected to recall facts associated with the Tweedie distribution, and to recognize the relationship between a reserve derived using a GLM versus the standard chain ladder approach.	

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Part a
Candidates were expected to recall the name of θ and $b(\theta)$ in the context of an EDF probability density function, or state how they affect an EDF distribution. Common mistakes included: • stating that they are the dispersion parameter • stating that they are the scale parameter
Part b
Candidates were expected to recall the restriction on the variance function of an EDF distribution that would give rise to a Tweedie distribution. Both the form of the variance function and the restriction on p must be identified. Specific results of the Tweedie distribution, such as $\mu = [(1 - p)\theta]^{\frac{1}{1-p}}$ were given credit. Responses that did not identify the correct restriction on p did not receive full credit. Common mistakes included: • not providing the restriction on p • confusing the concept of variance versus the variance function
Part c
Candidates were expected to recognize that a higher value of p will increase the variance of a Tweedie distribution and thus generate a distribution with a heavier tail. Alternatively, candidates can identify which member of a Tweedie distribution has a heavy tail, and how such a distribution would be generated by changing p. Responses that stated selecting a higher ϕ will lead to higher variance and thus a heavier tail did not receive credit. ϕ is a parameter of a distribution of the EDF family, so it does not affect the selection of a distribution. Common mistakes included: • stating decreasing p will increase tail heaviness • stating a specific value of p without clear justification why it would lead to a heavy tail
Part d
Candidates were expected to recognize when $p = 1$, the Tweedie distribution is an over dispersed Poisson (ODP) and it replicates the standard chain ladder method. Common mistakes include • using incorrect MLEs for the ODP cross-classified model • not stating the reason why the CL method applies